CSE-3001

SOFTWARE ENGINEERING

HOSTEL DIRECTORY AND MANAGEMENT SYSTEM

PROJECT REPORT



VIT®
Vellore Institute of Technology
(Deemed to be University under section 3 of UGC Act, 1956)

SCHOOL: SCOPE

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ABSTRACT

Hostels are known to pose as a home to a gigantic population of residents. This is what makes the management of such a complex an arduous task with a lot of room for inconsistencies. Many hostel management systems exist but a better software would be one that has more functionalities other than a resident directory.

The following report proposes one such system that not only contains a user directory access but also a mess feedback system, a medical emergency form submission option, a sports facility reservation scheduler and a live mess viewer.

Users can effectively find all these features in one convenient place. The sports facilities reservation scheduler serves the purpose of maintaining a definite schedule for access to prevent overcrowding and conflicts. Sometimes, the hostel may not be located right near a hospital which is why, during a medical emergency, immediate response from the hostel authorities to either call for an ambulance or provide medical attention is required. This is where the medical emergency is used. The authorities are immediately notified and appropriate response can be taken well before any critical danger is brought to a residents health. The mess feedback form is used to keep an accurate record of user reviews on the overall experience of a hostel mess.

The iterative waterfall model has been used to develop this software in various phases.

INTRODUCTION

With the increase of educational institutes in this country, there comes an increase in the number of residential accommodations. Hostels, to be more precise. Some institutes have large hostels with lakhs of students. Maintaining the records of these many residents, while being a difficult task, has still been implemented. However, it is very rare to find a software that also includes more functionalities.

A hostel's quality can only be kept on track if there is constant interaction, feedback and immediate responses that is maintained between the hostel and users. With critical feedback comes the chance to improve and overall maintain a harmonic environment. A large predicament in hostels, due to the large population, is overcrowding in the sports facilities such as gyms and sports courts. The only solution to solve such an issue is a scheduler that everybody follows. In times of medical emergency, immediate action needs to take place

Essentially, a software that facilitates the interaction between residents and the administration is required to provide a comfortable hostel experience. This is what the following software aims to achieve, keeping in mind the adverse effect of a faulty and badly managed hostel.

RELATED WORK

1. ***Magar, Shyamsundar & Jadhav, Mr & Said, Ms & Jadhav, Mr. (2021).*** **Hostel Management System and Aggregation.**
   This Hostel Management web application is mainly utilized to book hostel rooms, maintain a database of residents and update a community dashboard. The entire application is a WebApp that works both on IOS and Android. It has mainly been designed for hostel administrators.

2. ***Yaw, Mensah. (2022).*** **Hostel Management System Using Image Recognition. Engineering and Technology Journal. 07. 10.47191/etj/v7i7.09.**
   This hostel management system uses image recognition to formulate an item management system and allocates rooms to registered users. This system is a web application developed using C# programming language. This application includes a module that finds missing items, a module for hostel registration and also a module for giving feedback. In conclusion the model has made managing the affairs of the hostel easy.

3. ***Ayanlowo, K. & Shoewu, Engr. Dr. Oluwagbemiga & Olatinwo, S.O. & Omitola, Olusegun & Babalola, D.D.. (2014).*** **Development of an Automated Hostel Facility Management System. Journal of Science and Engineering. 5. 1-10.**
   In this paper, the development of an automated hostel accommodation management system is proposed. The codes for the automated system were developed using Visual Basic and Microsoft Access was used to develop the underlying database. It also has in-built authentication algorithm for preventing unauthorised access. This system mainly focuses on keeping a track of the hostel

residency history. It keeps track of the attendance history and also includes a feature that portrays the weekly mess bill so that users are aware of the payments to be made.

4. ***Prof. Deepali Narkhede, Rutuja Bamgude, Mayuri Sonawane, Mandar Shevade.*** **Hostel Management System (HMS)**

   This particular project deals with the problems on managing a hostel and avoids the problems which occur when carried manually. Identification of the drawbacks of the existing system leads to the designing of computerized system that will be compatible to the existing system with the system which is more user friendly and more GUI oriented. This project has the most basic functionalities that are required for a hostel management system with the added functionality of being able to view the hostel fee details.

5. ***Ritesh Kumar Bista, Aman Jung Karki, Beesu Venkat Mouneesh Reddy, Utkarsh Aakash, Dr. Rajasimha A Makaram, Shilpa Das.*** **Hostel Management System**

   This project is expected to limit human works and make hostel allocation much easier for student and hostel administrators with the help of the web application to hostel, naturally select the student from the waiting list and mess billing, outpass generation, complaint registration, and so forth. Students will get an endorsement notice in their mails. It advises guardians with respect to their wards and their presence in the hostel and their curricula will be informed to their parents using this model just in one touch.

REAL LIFE APPLICABILITY

In the past 10 years, anybody with access to technology has gotten adjusted to using WebApps or applications that have automated cumbersome tasks that used to be done manually before. Organizational applications, feedback applications and data transmission applications are now more in use than ever before.

In the perspective of a hostel community, the automation process is slowly becoming more popular because the standards of living are directly linked to our overall well-being, productivity and standings with respect to social strata. If the opportunity to make this aspect of human life better exists, steps should be taken to implement it at all costs.

Some of the main issues that hostellers face are standing in long mess lines, not being able to access sports facilities due to the lack of an organized slot occupation system and lack of access to a state of the art feedback system. The main objective of our project is to develop an easy to access website with functionalities that can potentially improve the quality of life for students.

With this technology, any hosteller can easily manage their own hostel life without any manual interruptions or facing any cumbersome processes.

It not only allows users and hostel administrators to keep a check on their activities but also provides an arena where each party is held accountable for not following the basic norms of existing.

INDIVIDUAL CONTRIBUTIONS

S Rajesh - 20BCE2755

- Database design and connectivity
- Registration portal
- Model structuring

Nida Ghani - 20BCE2850

- Front-end design
- Login portal
- Medical form

Yatish Sikka - 20BCE0896

- Mess feedback system
- Facility Registration
- Model structuring and planning

Anjan - 20BCE2767

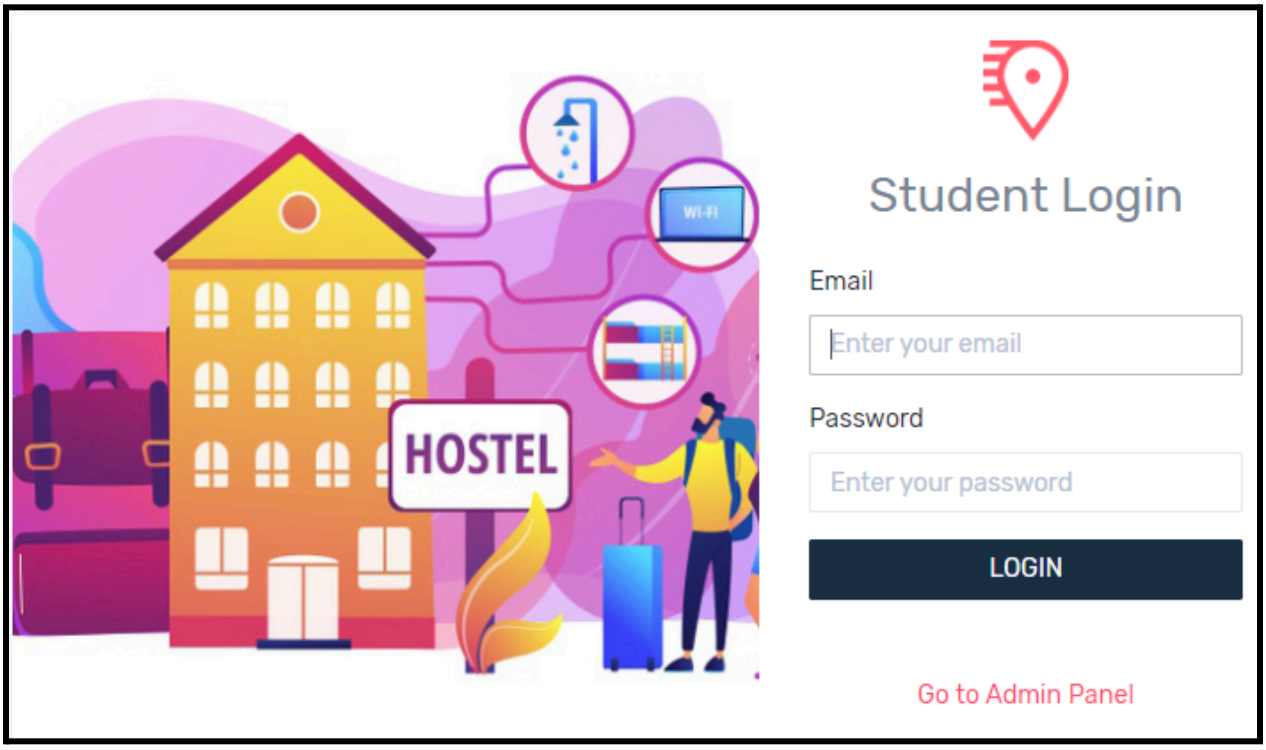
- Literature Survey
- Documentation

TOOLS AND TECHNOLOGIES USED

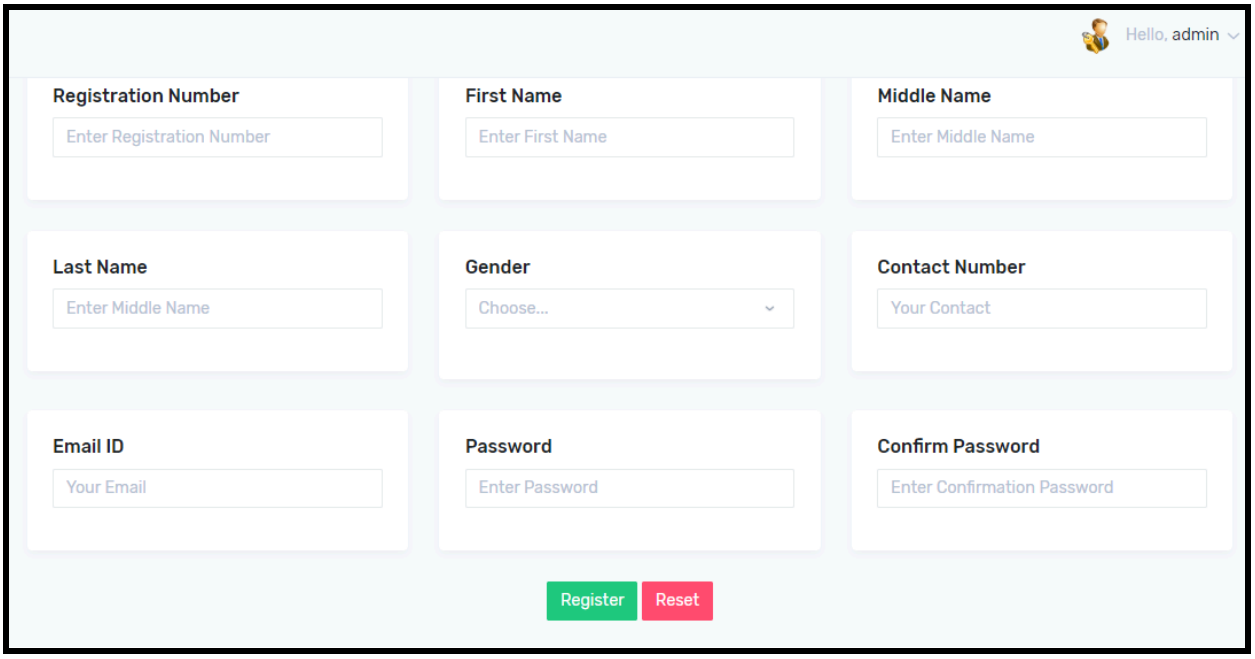
- PHP
- HTML
- CSS
- MySQL
- JAVASCRIPT
- BOOTSTRAP
- XAMPP
    1. Apache Server
    2. Localhost
- Windows 10 or above/ RAM- 4GB
- Browser - Google chrome/Microsoft Edge

PROPOSED SYSTEM PROCESS FLOW

1. **Login Page:**
   The user is first exposed to this page where they will login to the application with their respective credentials.

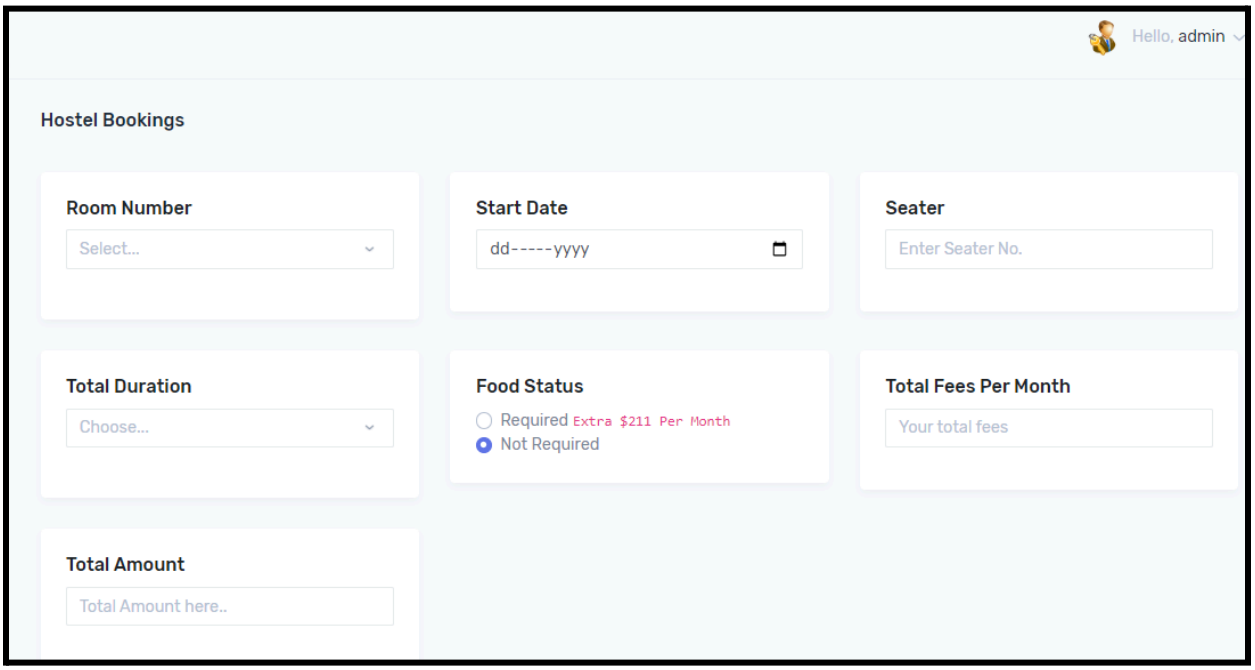


2. **Register resident:**
   If the user is new to the application, they can register to be a hostel resident using this feature. Their details will then be stored in a database for further use.

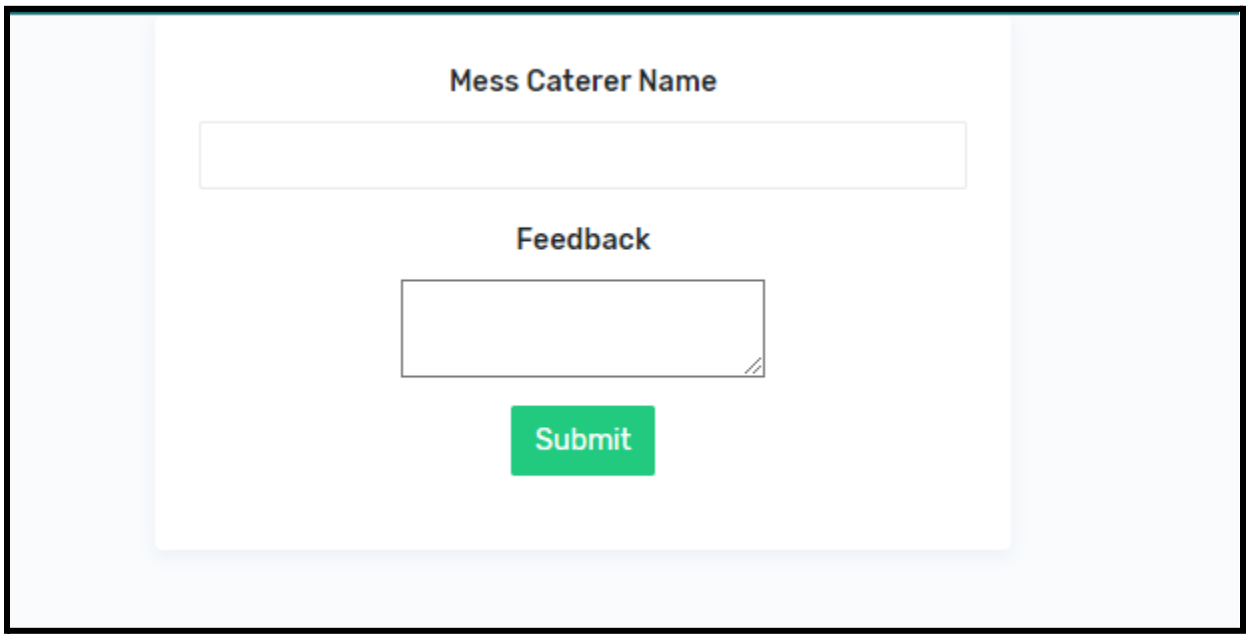


3. Book hostel:

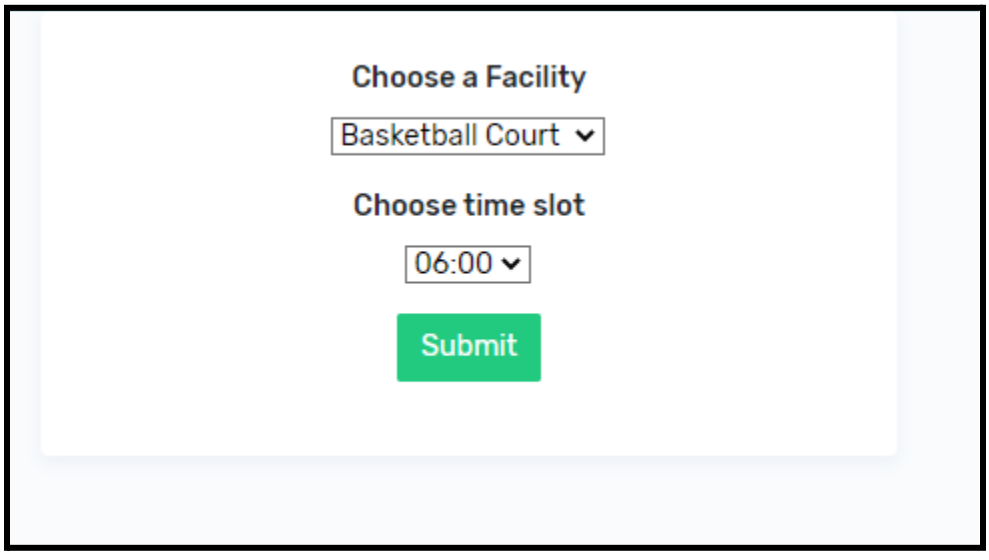
This option provides the user with different room types to choose from and effectively book the hostel.



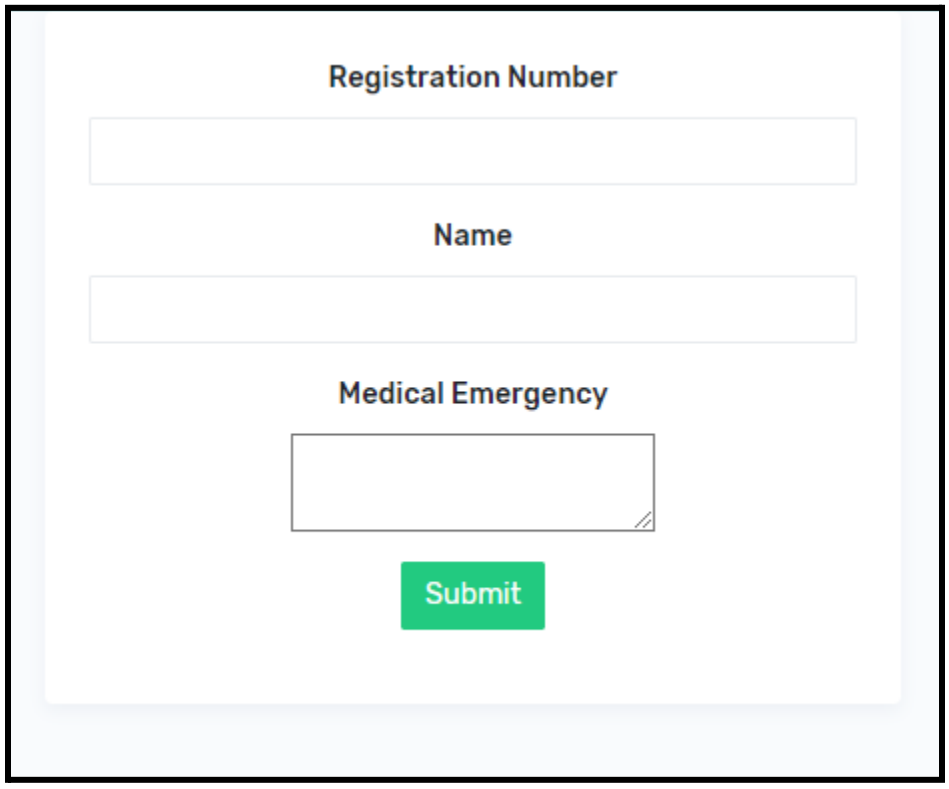
4. **Mess Feedback:**
   Any kind of mess feedback can be submitted to the hostel administrators with the help of this form. The feedback and user information is all stored in a separate database.



5. **Sports facilities reservation scheduler:**
   This functionality is used to book a slot in any of the hostel's sports facilities. It asks the user for the sports facility, the date and the time they want to book the facility for. It has a slot allocation system that books the users for a facility only for a defined time period.



6. **Medical emergency form submission:**
   If a resident is having a medical emergency and immediately needs to notify the authorities to arrange for any form of medical attention, they can submit their details with this form.



WORKING METHODOLOGY

In order to maintain a harmonious lifestyle in a hostel, there needs to be enough and quick interaction between the hostel administrators and residents. The functionalities we have included mainly consist of this one factor regardless of how much they may differ in design. The long lines, cumbersome paperwork and overall inefficiency of responses from the hostel authorities can be improved upon with the help of easy to access applications or websites.

1. **Project model: Iterative waterfall model**

   The iterative waterfall model provides feedback paths from every phase to its preceding phases, which is the main difference from the classical waterfall model. The iterative waterfall model will be suitable for our project because it is simple to understand and easy to use. It consumes less time for documentation and we can work more on developing the model. We also have feedback paths in this model which will help us get back to a point if we find any defect and correct it. The implementation portion of the project can potentially be tricky to carry out due to the 5 drastically varying types of functionalities. This is the reason why including feedback paths in the project model is imperative. We will constantly have the ability to review our work and adjust any errors that may arise.

2. **Work Breakdown structure:**

   In order to efficiently develop the website, each of our team members have provided their contributions at every stage. The following work breakdown structure showcases how the work has been divided and what time frame we have tried to stick to.

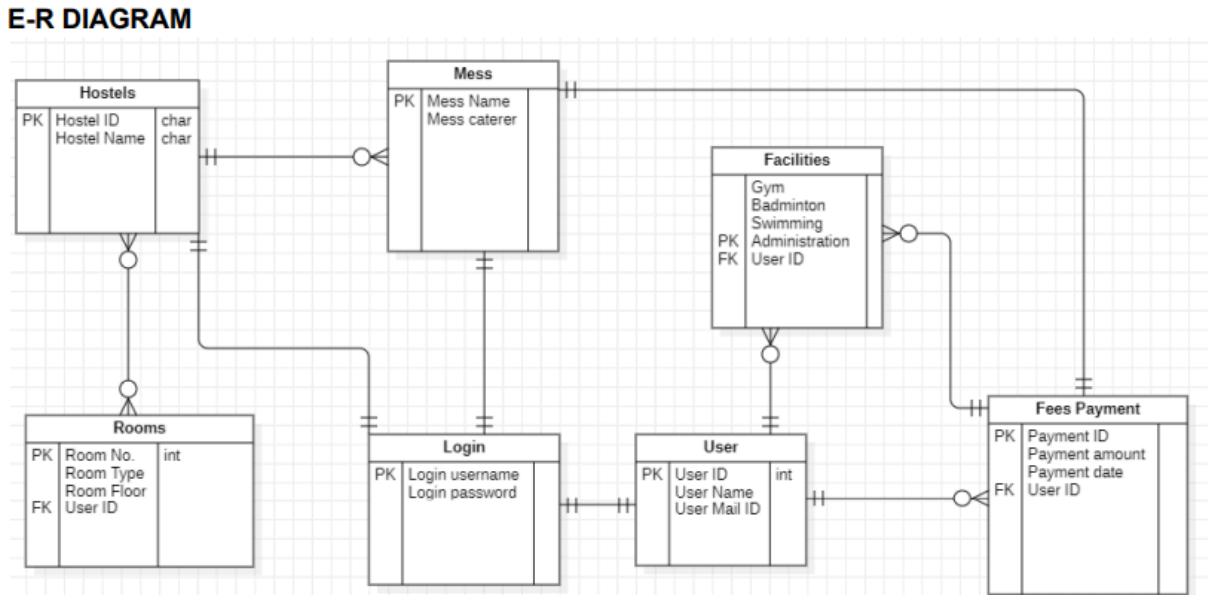
		Name	Duration	Start	Finish	Predecessors	Resource Names
1		Start	0 days	29/7/22 8:00 AM	29/7/22 8:00 AM		
2		**Feasibility Analysis**	1 day	**29/7/22 8:00 AM**	**29/7/22 5:00 PM**		
3		Technical Feasibility	0.5 days	29/7/22 8:00 AM	29/7/22 1:00 PM		Nida[50%];Yatish[50%]
4		Financial Feasibility	0.5 days	29/7/22 1:00 PM	29/7/22 5:00 PM	3	Rajesh[50%];Anjan [50%]
5		**Requirement Analysis**	18 days	**1/8/22 8:00 AM**	**24/8/22 5:00 PM**	**2**	
6		Market Research	4 days	1/8/22 8:00 AM	4/8/22 5:00 PM		Nida[50%]
7		Brainstorming	6 days	5/8/22 8:00 AM	12/8/22 5:00 PM	6	Rajesh[50%]
8		Feature List	8 days	15/8/22 8:00 AM	24/8/22 5:00 PM	7	Yatish[50%]
9		**Design**	10 days	**25/8/22 8:00 AM**	**7/9/22 5:00 PM**	**5**	
10		Feature prioritization	1 day	25/8/22 8:00 AM	25/8/22 5:00 PM		Nida;Rajesh;Yatish;Anjan
11		Architecture	4 days	26/8/22 8:00 AM	31/8/22 5:00 PM	10	Yatish
12		Specification Development	5 days	1/9/22 8:00 AM	7/9/22 5:00 PM	11	Nida
13		**Implementation**	28 days	**8/9/22 8:00 AM**	**17/10/22 5:00 PM**	**9**	
14		Developing Dashboard	2 days	8/9/22 8:00 AM	9/9/22 5:00 PM		Rajesh;Anjan ;Yatish;Nida
15		Developing UI	3 days	12/9/22 8:00 AM	14/9/22 5:00 PM	14	Anjan
16		Coding Hostel directory	4 days	15/9/22 8:00 AM	20/9/22 5:00 PM	15	Yatish
17		Coding Slot occupation system	4 days	21/9/22 8:00 AM	26/9/22 5:00 PM	16	Rajesh;Nida
18		Coding Live video feed	5 days	27/9/22 8:00 AM	3/10/22 5:00 PM	17	Nida
19		Coding Feedback system	3 days	4/10/22 8:00 AM	6/10/22 5:00 PM	18	Yatish;Anjan
20		Coding Medical Leave Form	3 days	7/10/22 8:00 AM	11/10/22 5:00 PM	19	Rajesh
21		Report Generation	4 days	12/10/22 8:00 AM	17/10/22 5:00 PM	20	Yatish
22		**Testing**	19 days	**18/10/22 8:00 AM**	**11/11/22 5:00 PM**	**13**	
23		Unit Testing	7 days	18/10/22 8:00 AM	26/10/22 5:00 PM		Anjan
24		Integration Testing	5 days	27/10/22 8:00 AM	2/11/22 5:00 PM	23	Nida
25		System Testing	2 days	3/11/22 8:00 AM	4/11/22 5:00 PM	24	Nida;Rajesh;Yatish;Anjan
26		User Testing	3 days	7/11/22 8:00 AM	9/11/22 5:00 PM	25	Rajesh
27		Deployment	2 days	10/11/22 8:00 AM	11/11/22 5:00 PM	26	Yatish
28		end	0 days	11/11/22 5:00 PM	11/11/22 5:00 PM	22	

3. Feature Specification :

The main functionalities of our project are:

- User registration
- Room registration
- Mess Feedback system
- Sports facilities reservation scheduler
- Medical Emergency form

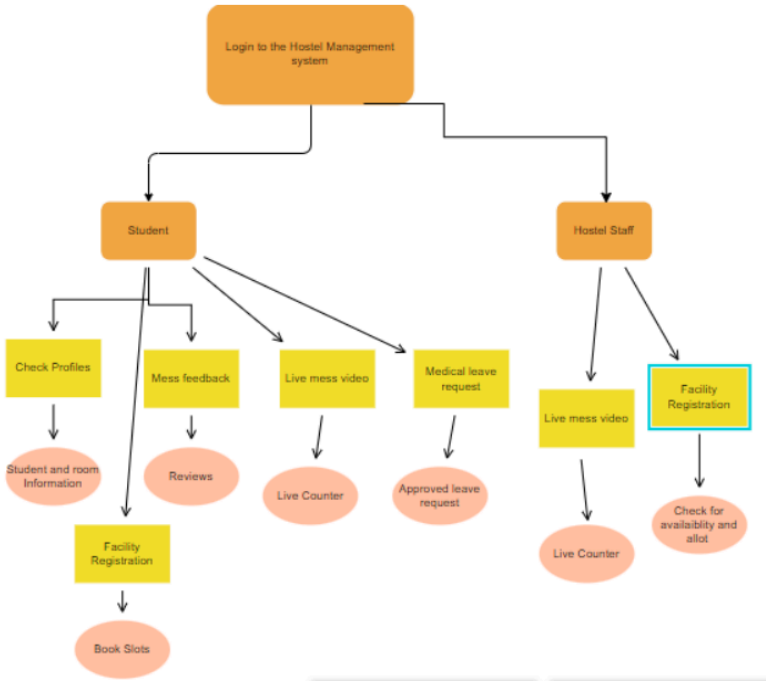
The interaction between these facilities can be portrayed with the following ER-diagram:



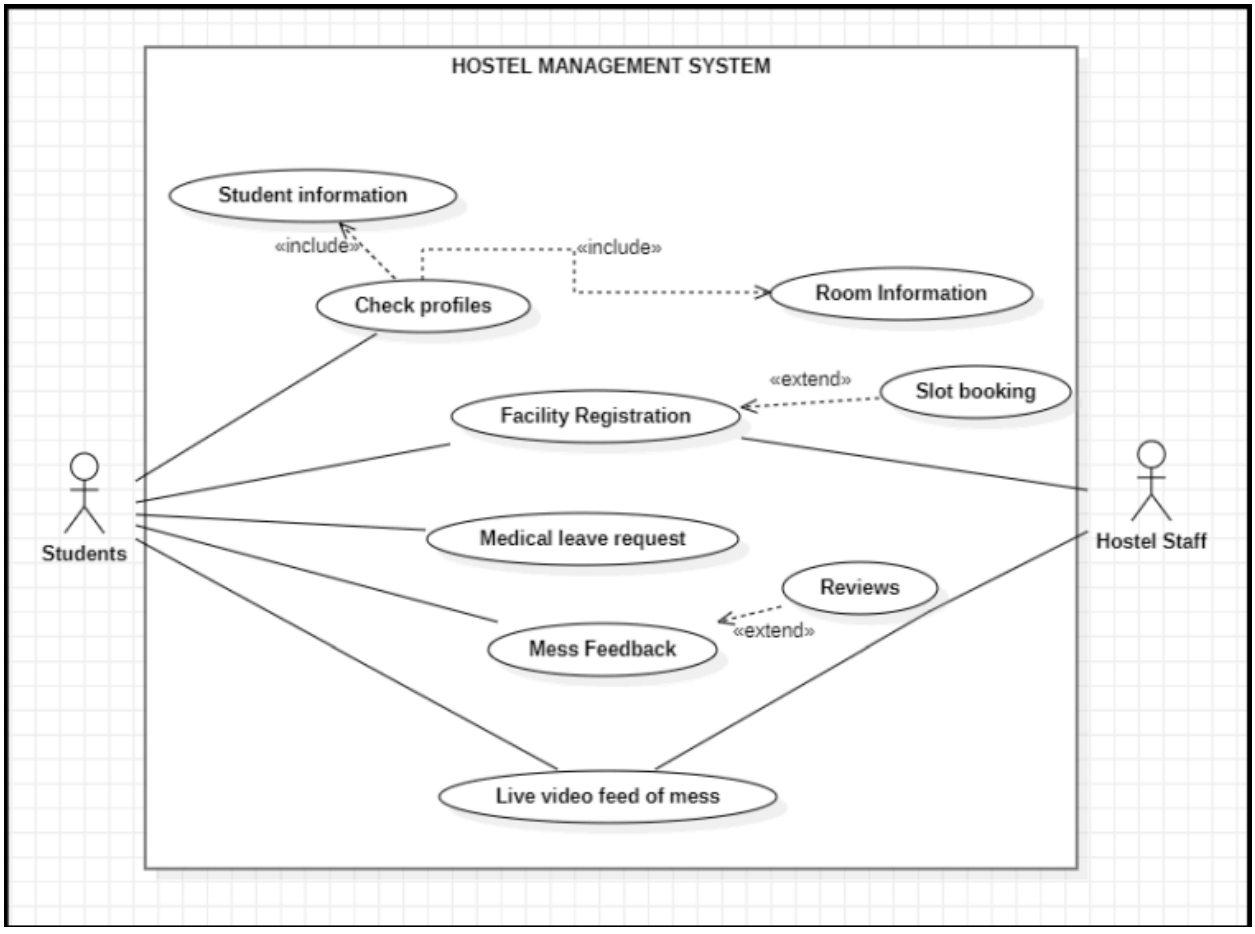
4. **Use case diagram and Structure chart:**

   The website has mainly been designed keeping in mind the interaction between the users and hostel administrators. The following diagrams portray how that interaction will take place.

Structure chart:



Use case diagram:



Student access:

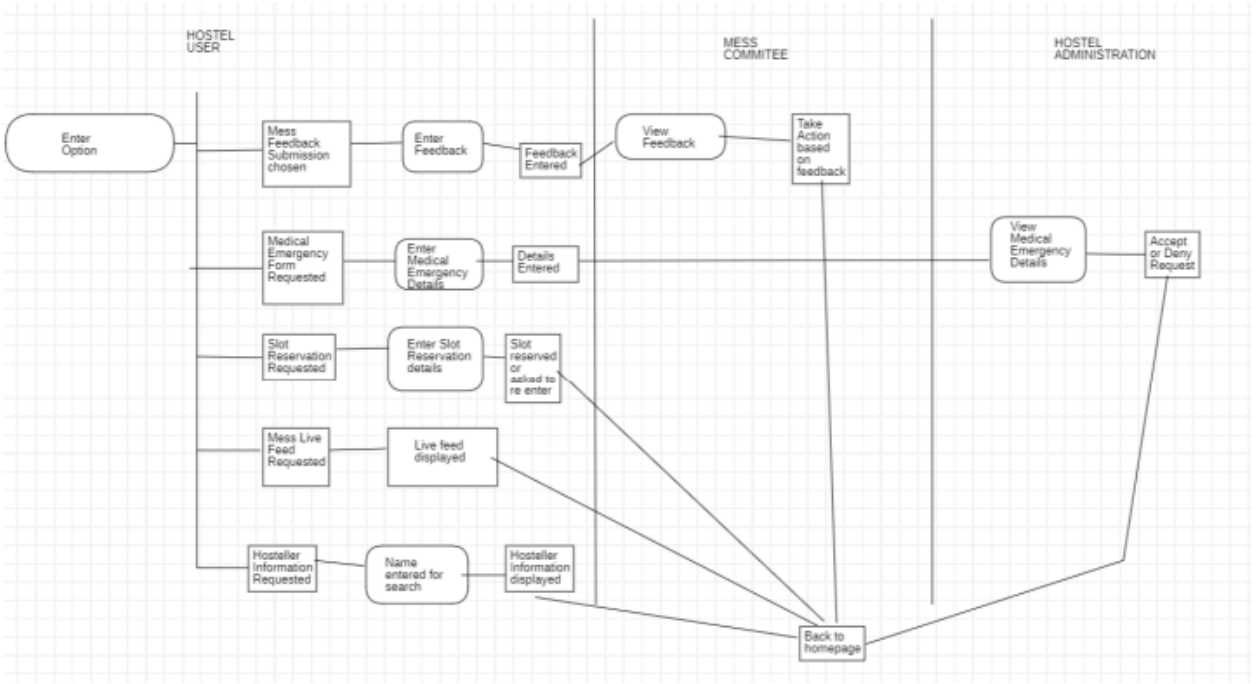
- Register profile
- Facility booking
- Medical emergency request
- Mess feedback

Hostel administrator access:

- Student database
- Feedback viewing
- Medical emergency approval

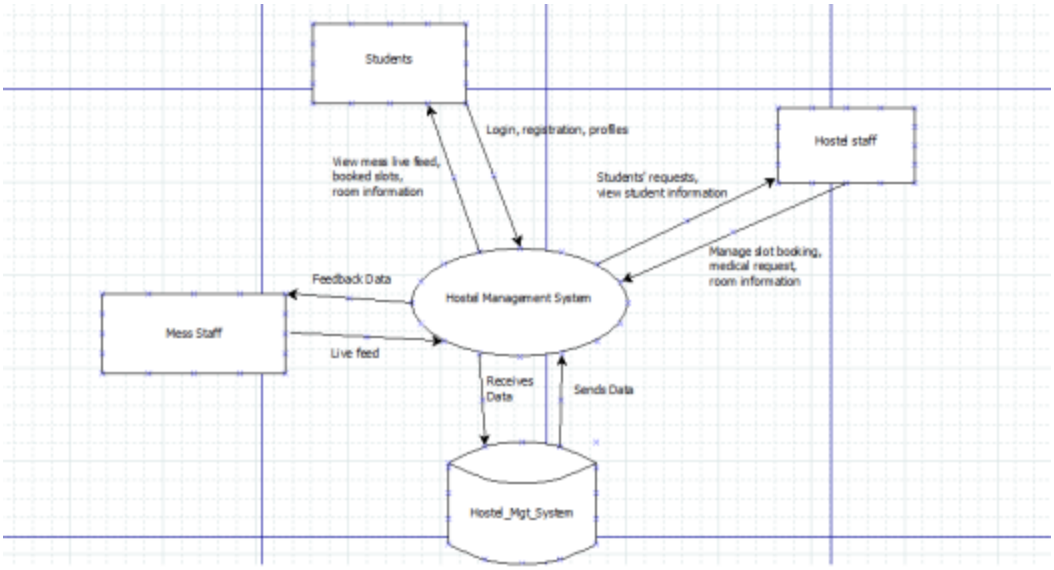
5. Activity flow diagram:

The general process flow for the entire system can be portrayed using the activity flow diagram:



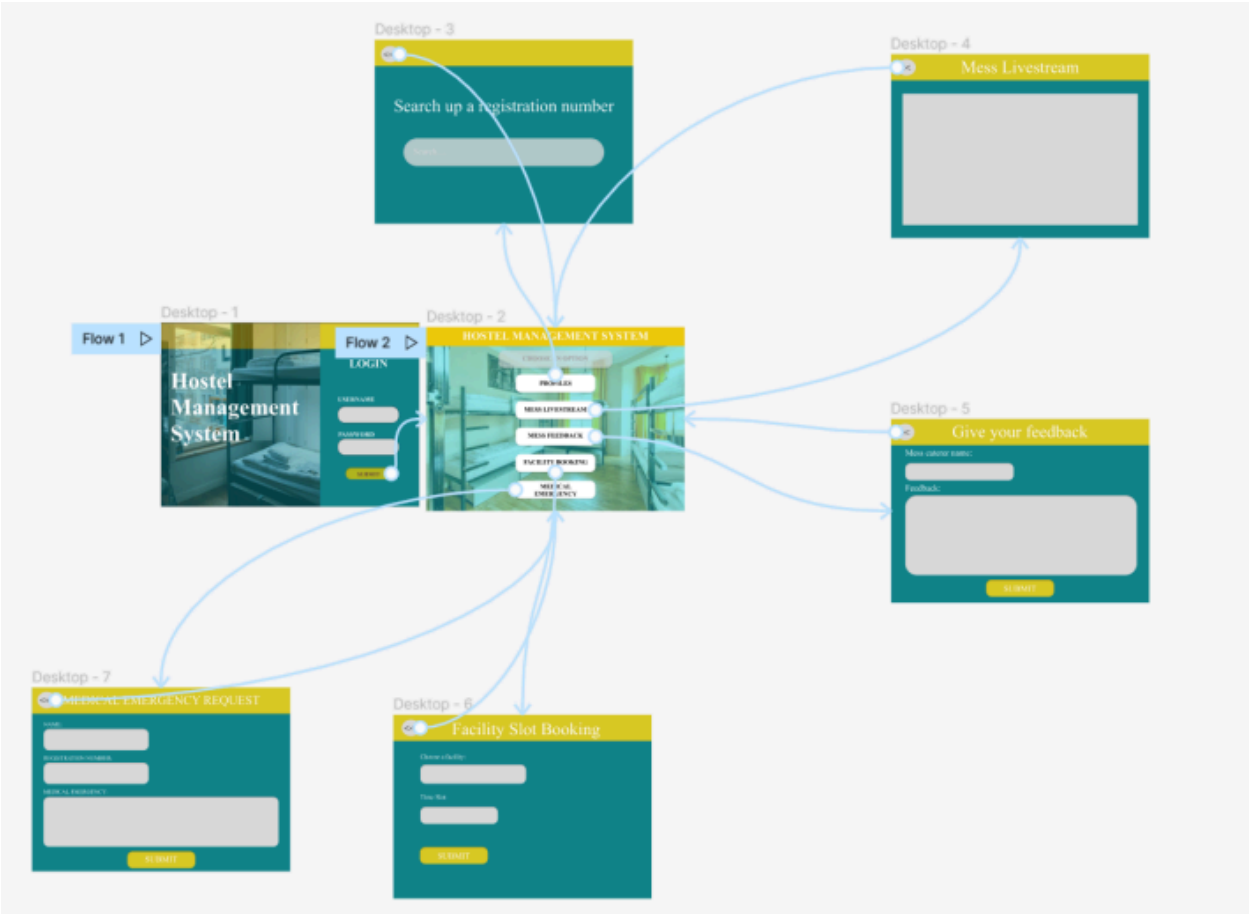
6. Architecture Diagram:

The architecture that has been followed can be seen in the following diagram:



7. User Interface:

While developing the website, we came up with a prototype that briefly describes what the final product will look like:


Desktop - 3
Search up a registration number
Desktop - 4
Mess Livestream
Desktop - 1
Flow 1
Hostel Management System
Flow 2
LOGIN
Desktop - 2
HOSTEL MANAGEMENT SYSTEM
PROFILES
MESS LIVESTREAM
MESS FEEDBACK
FACILITY BOOKING
MEDICAL EMERGENCY
Desktop - 5
Give your feedback
Feedback:
SUBMIT
Desktop - 7
MEDICAL EMERGENCY REQUEST
SUBMIT
Desktop - 6
Facility Slot Booking
SUBMIT

8. Risk assessment and risk mitigation:

The following chart describes the major risks involved with the website, their severity and category:

#	Risk Category	Score	Comments
	BUSINESS RISK	115	N/A
1	**Implementation complexity and impact to processes and business areas:**	60	
	- Implementation is not complex	10	The implementation is fairly complex with less prior knowledge
	- Implementation is complex, though team has prior experience	20	
	- Implementation is complex with many areas impacted and new methods introduced	30	
2	**Business Dependencies that might impact implementation:**	15	
	- No impact, No business dependencies exist	10	
	- Yes, some business dependencies exist but are low risk	5	
	- Yes, there are key business dependencies that might impact project implementation	0	
3	**Outages required during the implementation:**	40	
	- None required	20	
	- Outages to be planned after business hours	10	
	- Outages to be planned during business hours	10	
	TECHNOLOGY RISK	545	N/A
4	**Type of Project:**	195	
	- Maintenance (e.g., correct problems)	30	Maintenance might become an issue when the amount of data present exceeds the limit of the database and hence, data might be lost. While replacing, the data is at risk of being misplaced if appropriate care is not taken. When implementing a software package or allowing access to software by a third party, software is at risk of being penetrated and data might get tampered with.
	- Enhancement (e.g., add new features)	10	
	- Product introduction	20	
	- Process improvement/re-engineering	10	
	- Upgrade existing software or applications	15	
	- New development – replace existing system	30	
	- New development – develop system to support new business	20	
	- Implementation of software package in-house	30	
	- Outsourcing to external vendor	30	

5	**External Vendor Involvement:**	90	
	- No external or internal vendor required	0	An external vendor may be used to upgrade the software with the help of an expert.
	- No, an internal partner will be used for some project activites	30	
	- Yes, an existing external vendor will be used	10	
	- Yes, a known external vendor will be used but a new contract or PO is required	20	
	- Yes, and external vendor is required but not yet identified	30	
6	**Level of vendor support necessary for the technology after implementation:**	100	
	- N/A (no vendors involved)	10	Vendor support becomes necessary when the skill requirement of the software at hand exceeds those of the team members.
	- No subsequent vendor support necessary	20	
	- Minor support necessary (e.g., maintenance only)	30	
	- Major support necessary (e.g., programming, upgrades, etc.)	40	
7	**Newness of Technology (hardware, systems, databases, communication, etc.) used in the project:**	100	
	- N/A (no technology involved)	0	The software at hands implements already existing technology and utilises it to create innovative and new functionalities.
	- Technology is not new and has been proven or tested	20	
	- New to the business line	30	
	- New to University Services	30	
	- New to the education/learning services industry	20	
8	**Availability of Project Team Resources**	60	
	- We have the resources with the appropriate skill sets	20	The project requires external vendor help to refine and train the model.
	- We have resources but they will need to be trained	30	
	- The project will require external resources	10	

	IMPLEMENTATION RISK	150	N/A
	Will the Office for General Counsel or Purchasing be used during the negotiation, creation, or modification of the contract?	100	
	- No contract is required for project.	0	An official contract will have to be developed with the hostels that will make use of this application.
	- No, the existing contract will be used and no modifications or addendums will be required.	10	
	- Yes, the OGC and Purchasing will be highly involved with contract negotiation, creation, or modification.	20	
	- Yes, the OGC and Purchasing will be involved, but in a limited capacity	30	
	- No, the OGC and Purchasing will not involved with the contract negotiation, creation, or modification.	40	
10	**Has the Business Case been developed, reviewed and approved:**	50	
	- Yes, the Business Case/justification is documented and fully approved	0	The business case, strategy and deliverables are known at a fair level.
	- Yes, but there are some unknowns or assumptions the business case, stakeholders or product/service deliverables	30	
	- No, there are many unknowns or assumptions around the business case, stakeholders, strategy or product/service deliverables	20	
11	**Additional project risks not previously addressed [1]:**	0	
	-	0	
	-	0	
	-	0	
	TOTAL PROJECT RISK SCORE	810	**FINAL PROJECT RISK RATING: HIGH RISK**

Risk Mitgation and contingency plan:

#	Risk statement		(Scale 1-100%)	(Scale 1-10)	Exposure	Mitigation	Contingency	Triggers	Assignee
	Condition	Consequence	Probability	Impact					
1	Lack of skill to develop the software to its full potential.	The project will not be refined.	40%	7	7.5	Training the members in different skillsets with respect to the project.	Outsource the software development to third party vendors	Start of project	Program manager
2	Possible penetration into database from a malicious source.	Data deletion or alteration.	20%	10	9	Use of role level authentication that ensures access to only specific portions of the software.	Use of intrusion detection software to prevent denial of service attacks. Develop, test, and regularly review recovery procedures along with consistent backups.	Exposure of project to the marketplace.	IT manager, network manager
3	User Interface having an irregular or redundant data input flow.	Usibility decreases which may lead to low userbase.	30%	7	7	Software has to constantly be reviewed and team has to be on the lookout for new UI formats.	Updating the UI periodically to make it more user friendly.	When a user accesses the software.	Product Front end development team.
4	Lag in submitting the medical emergency form.	Serious medical consequences that could lead to a health hazard due to delayed medical attention.	10%	10	10	The database being used must reflect the changes in real time.	The hostel authorities must be notified with a notification.	When a medical emergency takes place and the person in danger requires permission to access immediate medical care.	Product development team.
5	Legitimacy of slots being booked.	Barrs the access to sports facilities when users enter the timeslot but does not show up.	50%	5	6	The authorities need to be alerted.	A slot updating and verification option should be available.	When a user books a slot but does not show up to use the facilities.	Product development team.

COMPARATIVE ANALYSIS

[1] This system mainly implements a hostel management system from the perspective of the administrator's ease of use. The residents utilize this system to mainly view the details of their profile. In comparison to our proposed model, this system has the added functionality of a community dashboard which serves a very useful purpose as all the hostel residents can simultaneously be alerted of any announcement.

[2] The unique feature of this model is that it uses image recognition to take an inventory of the personal belongings of hostel residents. Hence, in the case of a theft, the authorities can easily find the missing item. In comparison to our model, this added functionality keeps track of more than just the user information.

[3] This system is unique in terms of portraying the user's weekly mess bills and attendance history. In the perspective of our proposed model, these functionalities can serve as the basis for our project's future scope.

[4] This system has most of the basic functionalities such as user registration and room bookings. Our model is similar to this except we have three added functionalities such as mess feedback, medical emergency request and facilities reservation. The exceptional feature of this software is that it keeps a track of the hostel fees that are due.

[5] The following model is an advanced take on a hostel management system. It keeps track of the residents' whereabouts by using an RFID tag. This tag is scanned whenever the user enters the hostel or leaves it. It also has a feature that mails the guardians of the resident when they miss the attendance for a particular day. The mailing feature is quite innovative and can serve as the basis of future functionalities for our project.

CONCLUSION AND FUTURE SCOPE

The path to leading a good life involves good communication, a general understanding of basic norms and efficient scheduling of our activities. In a large university with lakhs of students, it can be challenging to maintain good well being aside from having to handle all forms of academic responsibilities. This is why having a fast and efficient system to communicate with the party that is in charge of your living quarters and store your details in one place is imperative.

The aforementioned system serves an incredibly useful purpose in a student's life. No longer would one have to wait for access to activities that should rightfully be available to them at a moment's notice.

The future holds many possibilities for this project. In addition to an alarm system for alerting administrators about student activity and inputs, we plan to add more features such as a community notice board, a private notes feature, a messaging platform that can be used between residents and a general announcement feature for the administrators.

In conclusion, the project works towards the betterment of hostel life and achieves to make the living dynamic more harmonious.

REFERENCES

1. *Magar, Shyamsundar & Jadhav, Mr & Said, Ms & Jadhav, Mr. (2021).* Hostel Management System and Aggregation.
2. *Yaw, Mensah. (2022).* Hostel Management System Using Image Recognition. Engineering and Technology Journal. 07. 10.47191/etj/v7i7.09.
3. *Ayanlowo, K. & Shoewu, Engr. Dr. Oluwagbemiga & Olatinwo, S.O. & Omitola, Olusegun & Babalola, D.D.. (2014).* Development of an Automated Hostel Facility Management System. Journal of Science and Engineering. 5. 1-10.
4. *Prof. Deepali Narkhede, Rutuja Bamgude, Mayuri Sonawane, Mandar Shevade.* Hostel Management System (HMS)
5. *Ritesh Kumar Bista, Aman Jung Karki, Beesu Venkat Mouneesh Reddy, Utkarsh Aakash, Dr. Rajasimha A Makaram, Shilpa Das.* Hostel Management System

APPENDIX

A. Link to PPT:

https://docs.google.com/presentation/d/1hIPQFCA8oHnyWJ2__8d6x4vryMi2kCL5OpNyrZVWGr8/edit?usp=sharing

B. Link to Pre-recorded Demonstration video:

https://drive.google.com/drive/folders/1VZXY12K5CYkz_-SMIUQ9WbHIQ1yrAz2U?usp=sharing

C. Link to access source files and file containing steps to execute the project:

https://drive.google.com/drive/folders/1rsIqsQKxmIjKAi9F5IJShJE3oRM-Lxzc?usp=share_link